Arista Access Node WHI-ACC-61 Whitefield East POP Routing Protocol Flap SOP

Node: WHI-ACC-61 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_9277

Description

The failure domain of Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) on the Arista Access Node WHI-ACC-61 at the Whitefield East POP can be attributed to a complex interplay of factors involving the interaction of multiple protocols and mechanisms. The Access Node, being a critical component of the network architecture, is responsible for facilitating the exchange of routing information between various domains. The BGP route churn, OSPF adjacency loss, and ISIS LSP flooding can be triggered by a variety of factors, including but not limited to, misconfiguration, hardware or software faults, and network topology changes. In the context of the Arista Access Node, the BGP routing protocol plays a pivotal role in exchanging routing information with neighboring routers. Any anomalies or irregularities in the BGP routing process can lead to route churn, resulting in the loss of OSPF adjacencies and subsequent ISIS LSP flooding. Therefore, it is essential to identify and address the root causes of this failure domain to prevent network instability and ensure optimal performance.

Pre-requisite Checks

- Verify that the remote hands are dispatched to the Whitefield East POP to facilitate any physical interventions.

- Check the status of the redundant links to ensure that they are functioning correctly.

- Confirm that the network topology and configuration are consistent with the expected design.

Potential Root Causes

Cause	Probability
BGP routing misconfiguration	High
Hardware or software faults	Medium
Network topology changes	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1, Description: This command displays a summary of the BGP routing table, providing information on the number of peers and the current table version. This can help identify any anomalies in the BGP routing process.

Command: show ospf neighbors, Expected Output: OSPF neighbor table is empty, Description: This command displays information on the OSPF neighbor relationships, allowing the identification of any lost
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adjacencies.

Command: show isis database, Expected Output: ISIS database is empty, Description: This command displays the ISIS LSP database, providing information on the current LSPs and their associated metrics. This can help identify any issues related to ISIS LSP flooding.
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Resolution Steps

Step 1: Step: Verify BGP routing configuration, Command: show bgp configuration, Description: This command displays the current BGP routing configuration, allowing the identification of any misconfigurations or anomalies.

Step 2: Step: Clear BGP routing table, Command: clear bgp *, Description: This command clears the BGP routing table, ensuring that all routing information is refreshed from the neighbors.

Step 3: Step: Verify OSPF adjacency relationships, Command: show ospf neighbors, Description: This command displays information on the OSPF neighbor relationships, allowing the identification of any lost adjacencies.

Step 4: Step: Reset ISIS LSP database, Command: clear isis database, Description: This command resets the ISIS LSP database, ensuring that all LSPs are refreshed from the neighbors.

Post-Check Verification

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Command: show bgp summary, Expected Output: BGP table version is 1, main routing table version is 1, Description: This command verifies that the BGP routing table is stable and consistent with the expected design.

Command: show ospf neighbors, Expected Output: OSPF neighbor table is not empty, Description: This command verifies that the OSPF neighbor relationships are stable and consistent with the expected design.

Command: show isis database, Expected Output: ISIS database is not empty, Description: This command verifies that the ISIS LSP database is stable and consistent with the expected design.
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Roll-back Procedures

- Step: Restore BGP routing configuration, Command: copy running-config startup-config, Description: This command restores the BGP routing configuration to the last saved configuration.

- Step: Re-establish OSPF adjacency relationships, Command: clear ospf adjacency, Description: This command re-establishes the OSPF adjacency relationships to their previous state.

- Step: Re-populate ISIS LSP database, Command: clear isis database, Description: This command re-populates the ISIS LSP database with the previous LSPs.